Geophys. J. Int. (1991) **107**, 191–196

RESEARCH NOTE

Palaeomagnetic directions (223–1.4 ka) recorded in the volcanites of Lipari, Aeolian Islands

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Accepted 1991 May 10. Received 1991 April 22; in original form 1990 September 10

SUMMARY

The island of Lipari is formed by Pleistocene volcanites emplaced during four main periods of volcanic activity. A study has been made of their magnetic properties, primarily with the aim of determining changes in the palaeomagnetic directions. Titanomagnetite is always the primary ferromagnetic mineral and its magnetic characteristics are common to the various lithotypes. However, the concentration of titanomagnetite and the degree of magnetic anisotropy vary systematically and correlate with the types of magma (basalt–andesite in the first and second volcanic periods; rhyolite in the third and fourth periods). All palaeomagnetic directions are of normal polarity (Brunhes epoch). Their mean overall palaeomagnetic pole (86°N, 238°E; $dp = 5°$, $dm = 6°$) is statistically indistinguishable from the geographic pole.

The variations in declination and inclination with age, however, are marked by some evidence of a discontinuity between 150 ± 10 and 127 ± 8 ka. This age can be correlated with the beginning of the Blake event of reverse polarity. The discontinuity might therefore correspond to magnetic excursions which occurred immediately before.

Key words: palaeomagnetism, Pleistocene, volcanites.

1 INTRODUCTION

The seven islands of the Aeolian archipelago are located in the Tyrrhenian sea near the northeastern coast of Sicily (Fig. 1). They are entirely composed of Pleistocene volcanic rocks. The earliest outcrops have K–Ar ages of about 500 ka (Gillot & Villari 1980). Two distinct stages of volcanic activity have been identified (Barberi *et al.* 1974). The first stage resulted in the formation of Panarea, Alicudi, Filicudi and part of Salina and Lipari, the second completed these two islands and gave rise to the still active volcanoes of Vulcano and Stromboli. A transition from calcalkaline to shoshonitic terms over the course of time has also been recognized (Barberi *et al.* 1973).

The volcanic activity in Lipari island took place in four successive periods (Pichler 1976, 1980) and lasted from 223 to 1.4 ka (De Rosa *et al.* 1985). The first two periods (223–42 ka) are characterized by prevailing basalt and basalt–andesite products, the third and fourth periods (42–1.4 ka) by rhyolite or alkali-feldspar rhyolite compositions. Chemical variations (mainly an increased potassium and silica content) are also reflected in different eruptive styles and flow morphologies. The first and second periods are marked by strombolian eruptions, the third and fourth by viscous lava flows and endogenous domes (Crisci *et al.* 1991).

Careful reconstruction of the stratigraphic sequences, supplemented by radiometric datings, makes the Lipari volcanites particularly suitable for studying the changes in the geomagnetic field over the last 223 kyr. Palaeomagnetic sampling was based on the 1:10 000 geological map published by Pichler (1976). The geochronological data were taken following the summary of the volcanic history of Lipari prepared by Crisci *et al.* (1991). Rock ages were obtained by fission-tracks (Bigazzi & Bonadonna 1973), ^{14}C on carbonized wood fragments (Crisci *et al.* 1981) and K–Ar (Gillot & Villari 1980).

Samples covering 12 volcanic units of the 15 main lava units recognized by Pichler (1976) were taken at 37 sites (Fig. 1), using an electric drill. Cores were oriented by magnetic and solar compasses. The present field declinations range from 2° to 10°.

2 MAGNETIC MEASUREMENTS

Laboratory procedures

The measurements were performed at the University of Turin's rock magnetism laboratory. NRM values were determined for all specimens by a JR-4 spinner magnetometer, and susceptibility and its anisotropy by a KLY-2 k-bridge. At least one specimen from each site was

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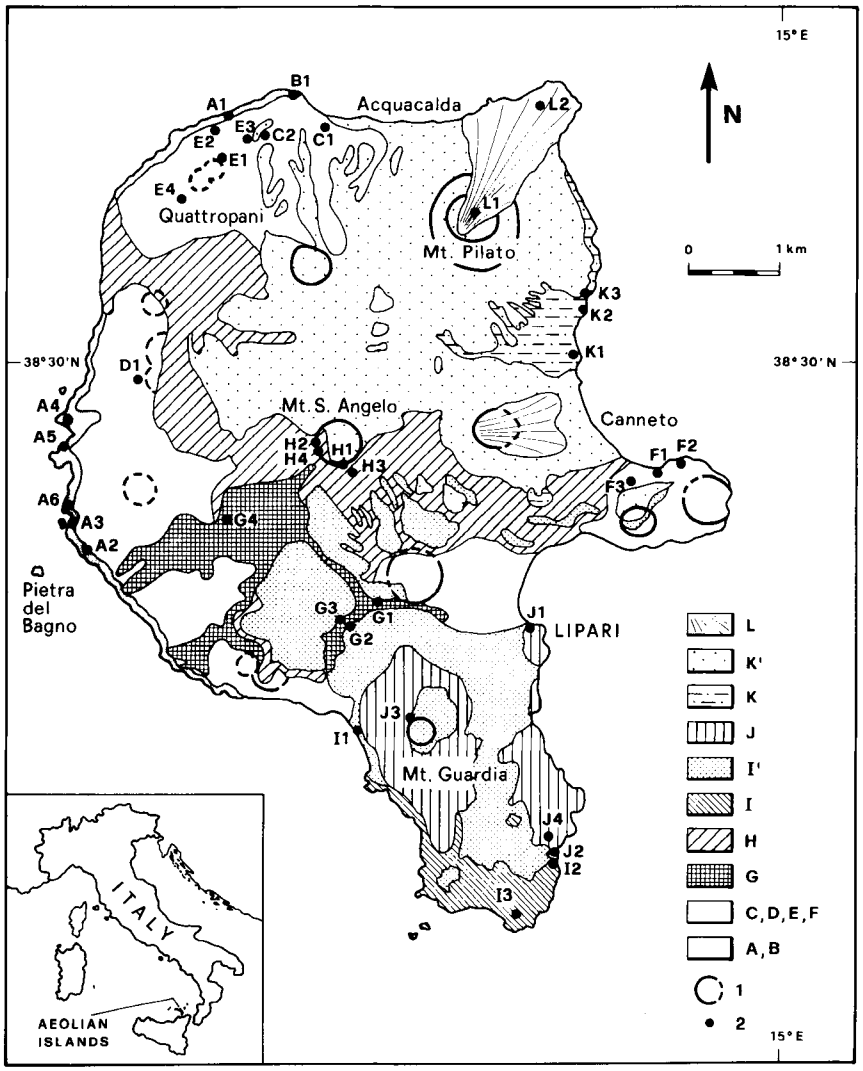


Figure 1. Geological map of the island of Lipari (after Pichler 1976, simplified). Letters: lithotypes, as specified in Table 1 (*I'* and *K'* = pumice series not sampled for this study). Numbers: 1 = crater rims, visible/probable; 2 = sampling sites.

subjected to thermal and AF demagnetization by Schonstedt and Molspin demagnetizers respectively, an IRM acquisition and back demagnetization cycle up to a maximum field value of 1.35 T by a Bussi pulse magnet, and a susceptibility versus temperature cycle by Bartington MSW2/F equipment.

Magnetic properties

The magnetic parameter values of the Lipari volcanites vary over a narrow range and point to a fairly uniform magnetic composition. Susceptibility gradually increases with temperature to a peak of about +50 per cent at 470°–500 °C, followed by a sharp fall to the Curie point, nearly always between 520° and 580 °C. The same temperature range is also apparent on the NRM demagnetization curves. The blocking temperature spectra may vary within lithotypes. NRMs, however, are entirely removed by 580 °C.

The IRM acquisition curves (Fig. 2) show that 90–95 per cent of saturation is reached in all lithotypes within an applied field value of 0.1 T, with the exception of the fourth volcanic period. Saturation is reached at about 0.2 T. The remanent coercive force needed to destroy the SIRM is of the order of 20–40 mT. The obsidian lavas of the fourth

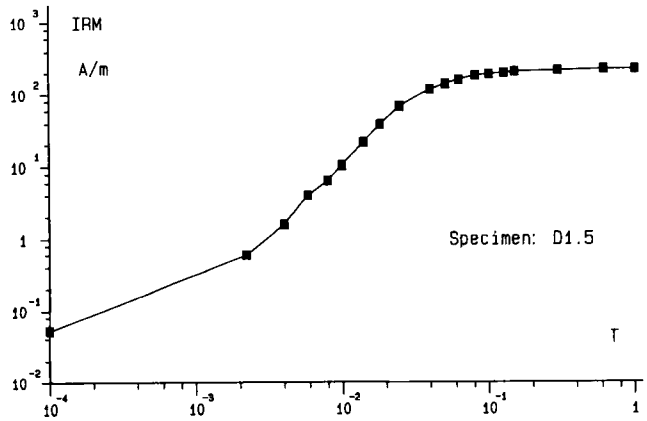


Figure 2. IRM acquisition curve.

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Table 1. Magnetic susceptibility and palaeomagnetic directions (ChRM, after AF treatment) of the Lipari volcanites. Symbols: N = number of specimens; k_m = mean susceptibility (SI units $\times 10^6$); P = degree of anisotropy (not reported for the pyroclastites); D, I = declination, inclination; α_{95} = semi-angle of confidence; PLA, PLO = latitude, longitude of the VGP.

PERIOD	LITHOLOGY	SITE	N	K_m	P	D	I	α_{95}	PLA	PLO
FOURTH PERIOD Chiefly Alkali rhyolites, minor rhyolites	L = Younger obsidian lava flows (Rocche Rosse)	L1	5	719	1.059	23.2	49.5	7.1		
		L2	10	1020	1.061	18.5	52.8	9.3		
		All	15	920	1.06	20.2	51.7	6.2	72	119
	K = Older obsidian lavas (chiefly of Mt. Pilato lava dome) and pumice fall deposits	K1	7	3980	1.026	0.1	59.3	13.9		
		K2	4	6411	1.189	28.3	70.8	7.7		
		K3	8	6944	1.012	356.3	60.3	3.7		
		All	19	5773	1.064	2.5	62.6	5.6	84	33
THIRD PERIOD Alkali rhyolites, to rhyolites	J = Younger endogenous domes (Mt. Guardia and Mt. Giardina)	J1	5	3683	1.098	7.5	49.7	6.1		
		J2	5	4582	1.079	359.3	50.4	5.0		
		J3	10	2159	1.062	4.3	57.6	3.6		
		J4	14	3027	1.106	17.1	31.6	9.1		
		All	34	3081	1.088	10.4	45.2	5.4	75	155
	I = Older endogenous domes and pumice fall deposits	I1	6	1721	=	350.6	54.0	4.9		
		I2	7	686	1.136	357.7	65.8	10.9		
		I3	10	1122	1.158	27.2	54.8	5.3		
		All	23	1145	1.150	10.6	59.2	5.6	82	91
SECOND PERIOD Chiefly quartz-latiandesites, minor quartz-latites, rhyodacites, dacites	H = Youngest lavas of Mt. S.Angelo volcano	H1	9	23353	1.015	341.6	37.0	12.5		
		H2	6	28223	1.016	344.6	49.2	1.9		
		H3	10	29119	1.018	315.5	45.9	20.2		
		H4	11	24420	1.017	347.8	42.8	19.1		
		All	36	26092	1.017	338.0	42.5	9.7	67	255
	G = Lavas with cordierite, andalusite, sillimanite and garnet	G1	8	4034	1.014	324.6	56.7	4.4		
		G2	10	12399	1.015	332.0	29.7	4.0		
		G3	7	7134	1.018	343.9	49.1	3.8		
		G4	10	4538	1.014	335.1	48.8	1.6		
		All	35	7188	1.015	333.7	44.8	3.7	65	264
FIRST PERIOD Chiefly quartz-latiandesites, minor quartz-andesites	F = Lavas of the Monterosa volcanoes	F1	5	24590	1.013	25.3	68.6	3.3		
		F2	13	27195	1.008	32.3	66.4	3.4		
		F3	7	29511	1.030	5.3	38.5	4.6		
		All	25	27297	1.012	19.5	59.8	5.7	75	91
	E = Lavas of Mt. Mazzacaruso-Chiesa Vecchia and welded scoriae deposits of Chiesa Vecchia	E1	4	3325	=	353.9	57.6	3.4		
		E2	4	3024	1.007	347.8	80.6	7.4		
		E3	5	1677	1.013	353.0	68.1	2.6		
		E4	7	26002	1.010	15.3	60.1	5.3		
		All	20	11514	1.010	2.3	66.1	4.5	80	24
	D = Aphanitic dikes of Timpone Pataso, Timpone Ospedale	D1	11	26567	1.006	344.5	36.5	2.5	67	236
	C = Lavas of Timpone Carrubo, Acquacalda, Mt. Chirica and deposits of chiefly scoriaceous ejecta of Timpone Pataso	C1	5	20262	1.008	338.5	57.4	3.8		
		C2	6	3920	1.016	343.1	60.2	9.2		
		All	11	11348	1.013	340.7	58.9	4.8	68	290
	B = Dikes cutting the submarine volcanics	B1	7	39707	1.022	345.5	46.0	17.2	86	132
	A = Submarine lavas and hyaloclastic products	A1	6	9152	1.025	7.7	60.6	3.6		
		A2	6	19316	1.005	18.1	48.1	3.7		
		A3	7	8953	1.013	359.5	55.1	3.7		
		A4	11	22306	1.011	4.2	56.4	1.9		
		A5	12	15584	1.025	351.2	54.8	15.7		
		A6	9	36584	1.012	3.1	52.6	2.0		
		All	51	19508	1.016	4.5	56.0	3.8	74	247

period show a slightly different pattern, requiring applied fields of 0.3 T for saturation. The remanent coercive force reaches 80 mT.

These results suggest that the primary ferromagnetic constituent of the Lipari volcanites is titanomagnetite, in agreement with the petrographic observations of Pichler (1980). The magnetic parameter values are characteristic of more or less pronounced oxidation of titanomagnetite, and of the occasional presence of haematite, in rocks clearly affected by secondary transformations (e.g. weathering or kaolinization).

Titanomagnetite is common to all lithotypes. Its content, on the other hand, varies systematically, as is shown by the mean susceptibility (k_m) values (Table 1). These are generally much higher for the first and second volcanic periods, than for the third and fourth periods. The degree of anisotropy ($P = k_{max}/k_{min}$), on the other hand, is always very low for the first two periods ($1.005 < P < 1.030$) and higher for the last two ($1.012 < P < 1.189$). Since titanomagnetite is the ferromagnetic mineral, the magnetic anisotropy is dependent on the shape and spatial orientation of the grains, so that a higher degree of anisotropy corresponds to a greater degree of preferred-orientation.

These systematic variations are correlated to the magmatic characteristics. The higher k_m values in the basalt–andesite rock types compared with the rhyolite reflect the general rule that magnetite contents increase with basicity. In addition, the greater viscosity of the rhyolite as opposed to that of the basalt–andesite magmas brings about enhanced magnetite crystal orientation, and hence more marked anisotropy.

Palaeomagnetic directions

The NRM intensities range between 0.5 and 7 A m^{-1}. A secondary magnetization component is nearly always present, as shown by analysis of the Zijderveld diagrams for the thermal and AF demagnetizations (Fig. 3). This component is randomly scattered within the NE and NW quadrants. It makes up less than 30 per cent of the NRM

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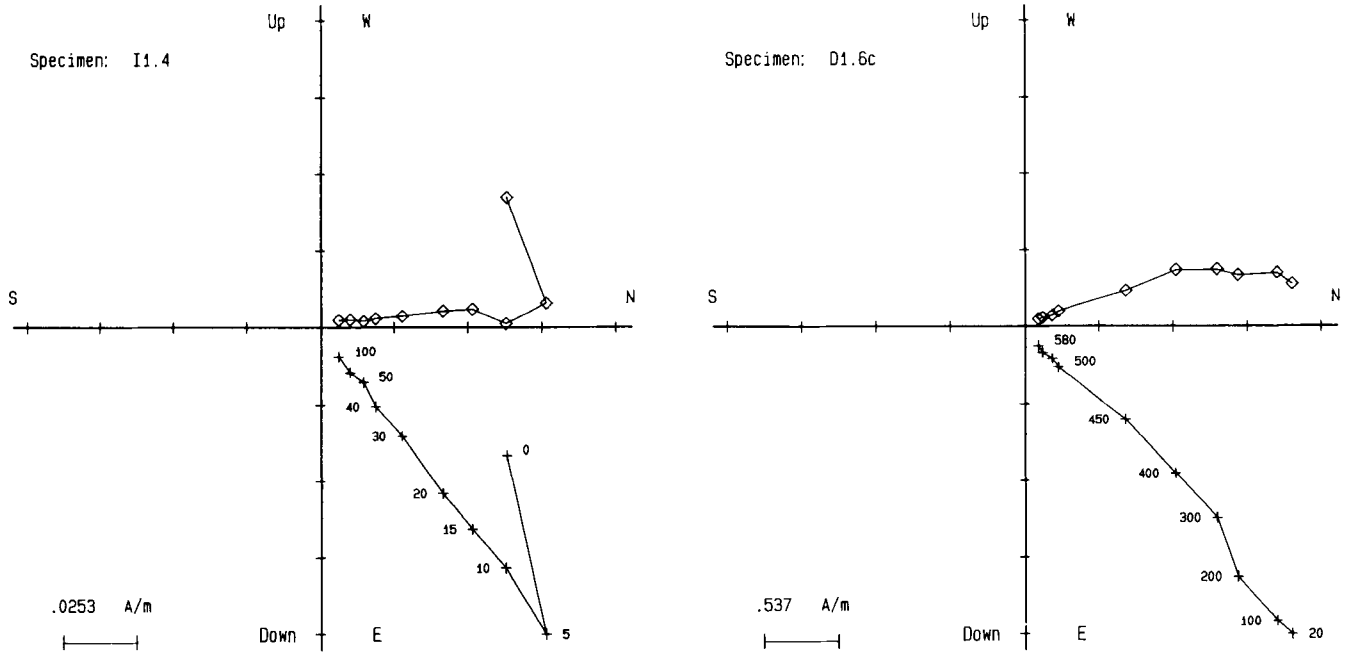


Figure 3. Zijderveld diagrams for AF and thermal demagnetization. Symbols: squares = declination; crosses = inclination; figures = H_{peak} values in mT, temperature values in °C.

and is removed during AF demagnetization at peak values of ≤20 mT, except for the obsidian lavas of the fourth period, which require fields of up to 40 mT. The secondary component resists thermal demagnetization up to varying temperatures, in some cases as high as 400 °C.

In view of its more uniform effect, AF demagnetization was chosen as the routine procedure for all specimens, and the identification of the stable component by Zijderveld diagrams has proven straightforward. This component is interpreted as the primary magnetization of the volcanites.

Table 1 reports the mean direction values per site and per lithotype, and the corresponding virtual geomagnetic pole (VGP). The palaeomagnetic specimens for each volcanic unit were collected from several distinct flows, whereas the datings are for single flows only. The mean directions per site thus represent a point-by-point sampling of the geomagnetic field, whereas the means per lithotype even out the variations that occurred over a period that cannot be precisely specified, but is in any event comparable with the uncertainty of the radiometric age. This does not apply to the oldest units A and B, whose duration cannot be assessed even hypothetically.

Direction values are also shown in the graphs in Fig. 4 showing the variations in declination and inclination as a function of the ages of the rocks. The vertical and horizontal sides of the shaded area surrounding the mean lithotype point represent the age uncertainty and the semi-angle of confidence α_{95}, respectively. This area is not shown for undated lithotypes B, D, E, F, G. Their position along the time axis is approximate, and has been assigned according to the stratigraphic sequence and the dating of the other rocks.

3 DISCUSSION

The results of this study show that the Lipari volcanites carry normal polarity magnetizations consistent with eruption during the normal polarity Brunhes chron. The overall mean direction of magnetization and the values of the relevant statistical parameters are: $D = 356.6°$, $I = 54.9°$, $N = 37$, $R = 35.73$, $k = 28$, $\alpha_{95} = 4.5°$, where R is the vector sum of the N unit vectors, k is the precision and α_{95} the semi-angle of confidence.

The spread of the directions around the mean is comparable to that for other coeval volcanic regions, such as the Central Massif in France (Doell 1970). Doell's formulae give a total angular standard deviation $s = 15.7°$. If account is taken of the experimental errors, this falls to $s = 15.3°$.

The mean direction of the Lipari volcanites is not statistically distinguishable from the axial dipole direction for the position of the island. The angle between the two directions, in fact, is only 3.5°. The same is true of the palaeomagnetic pole, whose position is: Lat. N = 86°, Long. E = 238°, $dp = 5°$, $dm = 6°$, where dp, dm define the 95 per cent confidence oval.

The Lipari volcanites have not provided any evidence of the reverse directions referable to polarity subchrons (such as the Laschamp, Mono Lake, Blake) reported by other workers for Brunhes volcanites and sediments. The Blake event, in particular, has been identified from sea bed sediment cores in two Thyrrenian Sea sites. Creer, Readman & Jacobs (1980) observed it in clays from the Gioia Tauro coring site, near the Calabrian coast, and assigned it an age between 155 and 105 ka. In the KET 8004 site, a few hundred kilometres to the NW of the Aeolian Islands, it was identified by Tucholka *et al.* (1987) and dated between 116 and 108 ka. Thouveny, Creer & Blunk (1990) have tentatively interpreted a reverse-to-normal transition recorded in the sediments of the Lac du Bouchet (Massif Central, France), and just younger than 117 ka, as the end of the Blake event.

Examination of Fig. 4 shows that the fairly regular patterns of the declination and inclination curves display some evidence of a discontinuity related to the lithotypes D,

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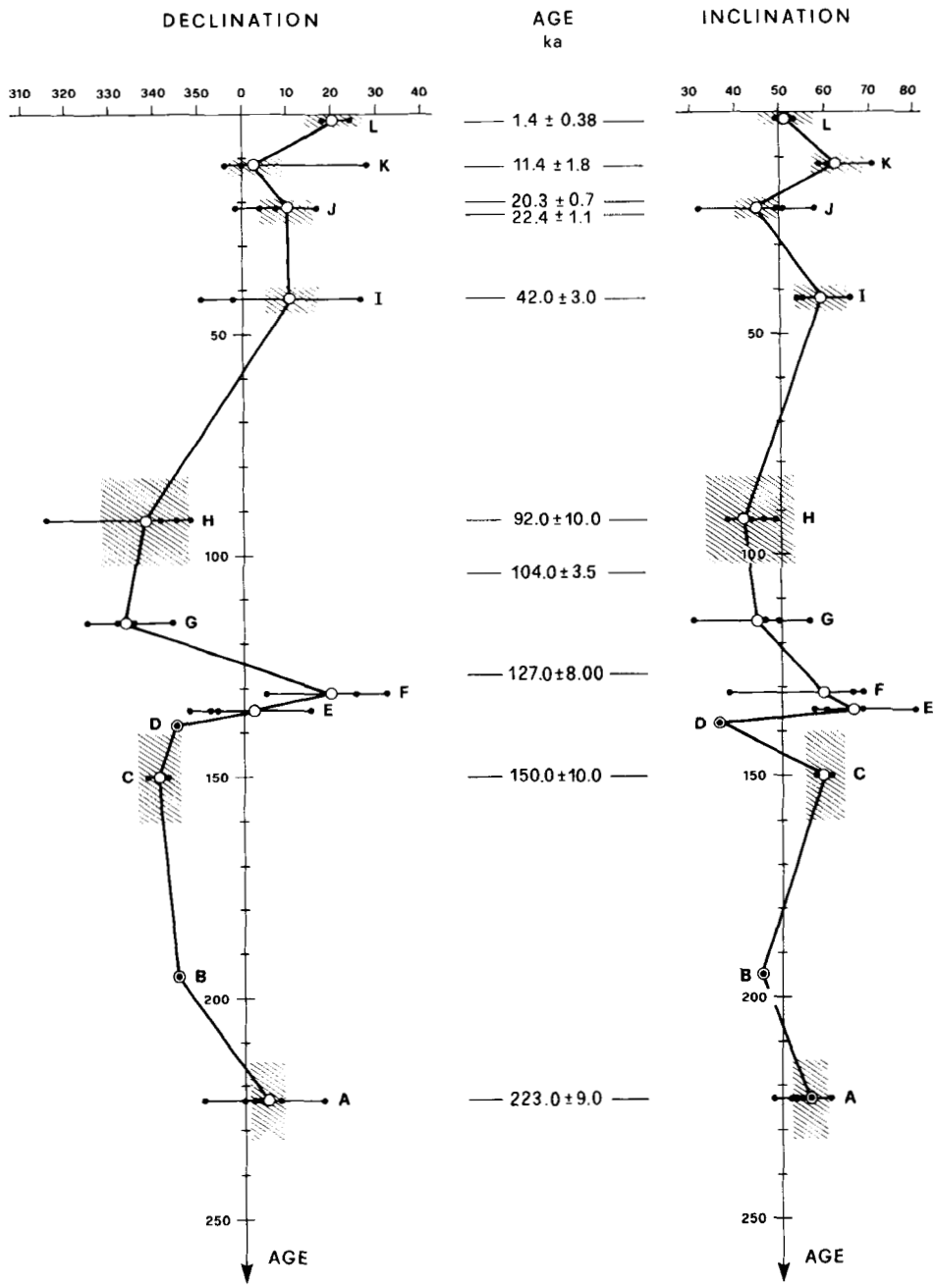


Figure 4. Variations of palaeomagnetic direction (ChRM, after AF treatment) in function of age of the Lipari volcanites. Symbols: points = site means; circles = lithotype means. The sides of the shaded areas are equal to the age uncertainties and to the values of the semi-angles of confidence α_{95}.

E and F. These are not dated, but their stratigraphic position is tightly constrained between 150 ± 10 and 127 ± 8 ka. The agreement between the various age values suggests that this discontinuity might correspond to geomagnetic excursions immediately before the Blake event. As the matter now stands, this is a working hypothesis that we intend to check by extending our palaeomagnetic sampling to include the other volcanoes in the Aeolian Islands.

ACKNOWLEDGMENTS

Financial support for this work was received from the Italian Ministry of Public Education (60 per cent funds). The authors wish to express their thanks to M. Miletto for her assistance during the field work, to G. M. Crisci and colleagues for a preprint of their paper and to M. Manzoni and two anonymous referees for the improvement of the manuscript.

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REFERENCES

Barberi, F., Gasparini, P., Innocenti, F. & Villari, L., 1973. Volcanism of the Southern Tyrrhenian Sea and its geodynamic implications, *J. geophys. Res.*, **78,** 5221–5232.

Barberi, F., Innocenti, F., Ferrara, G., Keller, J. & Villari, L., 1974. Evolution of Aeolian Arc volcanism (Southern Tyrrhenian Sea), *Earth planet. Sci. Lett.*, **21,** 269–276.

Bigazzi, G. & Bonadonna, F., 1973. Fission track dating of the obsidian of Lipari Island (Italy), *Nature,* **242,** 322–323.

Creer, K. M., Readman, P. W. & Jacobs, A. M., 1980. Palaeomagnetic and paleontological dating of a section at Gioia Tauro, Italy: Identification of the Blake event, *Earth planet. Sci. Lett.*, **50,** 289–300.

Crisci, G. M., De Rosa, R., Esperança, S., Mazzuoli, R. & Sonnino, M., 1991. Temporal evolution of a three component volcanological system: the Island of Lipari (Aeolian Arc, Southern Italy), *Bull. Volcanol.*, **53,** 245–260.

Crisci, G. M., De Rosa, R., Lanzafame, G., Mazzuoli, R., Sheridan, M. F. & Zuffa, G. G., 1981. Monte Guardia Sequence: a Late-Pleistocene eruptive cycle on Lipari (Italy), *Bull. Volcanol.*, **44–3,** 241–255.

De Rosa, R., Gillot, P. Y., Lanzafame, G. & Mazzuoli, R., 1985. The Island of Lipari, IAVCEI Scientific Assembly, *Excursion Guidebook*, pp. 114–124, Catania.

Doell, R. R., 1970. Paleomagnetic secular variation study of lavas from the Massif Central, France, *Earth planet. Sci. Lett.*, **8,** 352–362.

Gillot, P. Y. & Villari, L., 1980. K/Ar geochronological data on the Aeolian Arc volcanism. A preliminary report, *Open File Report 3/80*, CNR IIV.

Pichler, H., 1976. *Carta Geologica dell'Isola di Lipari (Scala 1:10 000)*, Litografia Artistica Cartografica, Firenze.

Pichler, H., 1980. The Island of Lipari, *Rend. Soc. It. Min. Petr.*, **36,** 415–440.

Thouveny, N., Creer, K. M. & Blunk, I., 1990. Extension of the Lac du Bouchet palaeomagnetic record over the last 120 000 years, *Earth planet. Sci. Lett.*, **97,** 140–161.

Tucholka, P., Fontugne, M., Guichard, F. & Paterne, M., 1987. The Blake magnetic polarity episode in cores from the Mediterranean Sea, *Earth planet. Sci. Lett.*, **86,** 320–326.

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